

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

D. Range Xor Subsequence Query

time limit per test: 2 seconds
memory limit per test: 256 megabytes

You are given a sequence of n integers $a_1, \dots, a_n$. Your task is to process the q queries of the following:

- Given l, r, x , check if it's possible to choose several integers from $a_l, a_{l+1}, \dots, a_r$ such that their bitwise XOR is x . Print "Yes" if possible, print "No" otherwise.

Input

The first line of input contains a single integer n ($1 \leq n \leq 10^5$).

The second line contains n integers $a_1, \dots, a_n$ ($0 \leq a_i < 2^{60}$) denoting the sequence.

Then the next line contains a single integer q ($1 \leq q \leq 5 \times 10^5$).

Then, for the following q lines, each line contains 3 integers l, r, x ($1 \leq l \leq r \leq n, 0 \leq x < 2^{60}$) denoting a query.

Output

For each query, print a single line of 'Yes' or 'No'.

Example

input	Copy
5 2 4 1 16 8 4 1 3 20 2 4 20 1 5 31 1 5 32	
output	Copy
No Yes Yes No	

Introductory Problems: XOR Basis

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
392887776	Oct/02/2026 01:49	Accepted
392887739	Oct/02/2026 01:47	Wrong answer on test 1
392887724	Oct/02/2026 01:47	Compilation error
325939486	Jun/25/2025 01:39	Wrong answer on test 5
325939369	Jun/25/2025 01:36	Wrong answer on test 5
325939336	Jun/25/2025 01:34	Wrong answer on test 5
325900628	Jun/24/2025 19:13	Memory limit exceeded on test 13
325899925	Jun/24/2025 19:08	Memory limit exceeded on test 11
325899426	Jun/24/2025 19:04	Memory limit exceeded on test 11
325898428	Jun/24/2025 18:58	Time limit exceeded on test 11